Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Michael Grosshauser	Email	Michael.grosshauser@lbf.fraunhofer.de
Institute Name	Fraunhofer Institute for Structural Durability and System Reliability LBF			
Address	Schlossgartenstraße 6, 64289 Darmstadt, Germany			

2. Technology Overview

Title of Proposed Technology	*Oxyimide-based, halogen-free flame retardants (including phosphorus-functional variants)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Mobility (Chemicals)		*Advanced Bio (Chemicals)*	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	*Special Functional Materials Technology*	*Fine Chemicals*	*Chemical/Bio Functional Materials*
Concept of Proposed Technology	*Core concept:* • *Develop halogen-free flame retardant (FR) packages centered on organic oxyimides (mono-/oligo-/polymeric) and phosphorus-functional oxyimides (phosphite/phosphonite/phosphate). Combine with aluminum/zinc phosphinates and related P-FRs to achieve high flame performance at reduced total loading, low migration, and stable processing/mechanics.* *Main functions:* • *Dual-phase FR action: condensed-phase char promotion + gas-phase radical quenching via P-species.* • *Low migration durability: polymeric/oligomeric oxyimides minimize blooming, fogging, and VOC.* • *Processing robustness: color-neutral options; stable rheology; recyclability-friendly behavior.*			

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	Potential uses: • *Electrical/electronic (E&E) housings/connectors, white goods, automotive interior/exterior parts, rail/building components (subject to codes), and appliance components requiring UL/IEC compliance.*
Rationale for Joint R&D Project	*Industry need:* • *Replace halogenated systems to meet RoHS/WEEE and ESG targets, reduce smoke/corrosivity, and avoid antimony. Achieve UL 94 ratings at lower total loading while retaining mechanicals/color and minimizing emissions and migration.* *Why collaboration:* • *Research institute (Fraunhofer LBF): IP and know-how on oxyimides and P-functional oxyimides; polymeric low-migration architectures; FR formulation science; analytics (UL/LOI/cone, TGA/DSC/TGA-MS/FTIR), migration/VOC/fog, durability, and LCA/ESG.* • *Korean SMEs: fine-chemical synthesis and scale-up, masterbatch/compounding, pilot part fabrication, local conformity testing and path-to-market. Joint work accelerates scale-up, certification, and adoption in Korean Advanced Mobility supply chains.*
Key Words	1. *Oxyimide flame retardants* 2. *Phosphorus-functional oxyimides* 3. *Low-migration polymeric FR* 4. *Sb-/halogen-free FR packages* 5. *Phosphinate synergy*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5-6)	End of the Project	Level (7)
R&D Objectives	*Oxyimide families:* • *Mono-/oligo-/polymeric oxyimides optimized for low volatility/migration and color neutrality.* • *Phosphorus-functional oxyimides (phosphite/phosphonite/phosphate) enabling dual FR + stabilization.* *Formulation modules:* • *Synergy with Al/Zn phosphinates/-ites/-ones; nitrogen donors as needed; nucleation/anti-drip tuning.* • *Masterbatch delivery and dispersion control across PP, PA6/66, PBT, PC/ABS, ABS/HIPS.* *Low-emission designs:* • *Oligomeric/polymeric architectures; optional microencapsulation; migration/VOC/fog compliance.* *Scale-up and compliance toolset:* • *REACH/K-REACH-ready data, color control, processing windows, and recycling/reprocessing stability.*			
R&D Target Specifications	*Flammability (representative targets by polymer):* • *PA6/PA66, PBT: UL 94 V-0 at 1.6 mm.* • *PC/ABS: UL 94 V-0 at 1.6 mm (V-0 at 3.2 mm with reduced total loading).* • *PP (select grades): V-2 at 1.6–3.2 mm; V-0 at 3.2 mm for target formulations.* *LOI (ISO 4589-2): ≥28 vol% for PA/PBT; ≥26 vol% for PC/ABS; ≥23 vol% for PP targets.* *Cone calorimetry (ISO 5660, 35–50 kW/m²):* • *PHRR reduction ≥30% vs halogen-free P-FR baseline at comparable rating.* • *SEA (smoke) reduction ≥20%; THR reduction ≥15%.* *Durability/processing:* • *Mechanical retention ≥90% of non-FR baseline (E-modulus/impact) at target loadings.* • *ΔMVR ≤10% after 3x re-extrusions (defined shear/thermal history).* *Emissions/migration/color:* • *VOC (VDA 278) ≤200 µg/g; Fog (DIN 75201-B) ≤3 mg; extractables ≤0.5% (7 d/40°C in application-relevant media).* • *Color ΔE* ≤1.0 after 1000 h heat/UV exposure (application-appropriate).* *Compliance:* • *Halogens not intentionally added; antimony-free; RoHS/WEEE alignment.*			
R&D Plan	*Approach* • *Mechanism-led design of oxyimide and P-functional oxyimide FRs emphasizing low migration and synergy with phosphinates to lower total loading and smoke while maintaining processing/mechanics. Iterative screening → optimization → pilot scale → application certification.* *Methodologies*			

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- *Analytics: TGA/DSC/TGA-MS, FTIR, rheology (MFR/MVR), DMTA; migration & extractables; VOC (VDA 278) and fog (DIN 75201-B).*
- *Fire testing: UL 94 (IEC 60695-11-10), LOI/(N)OI, ISO 5660 cone calorimetry; drip and char analytics.*
- *Processing: twin-screw compounding, masterbatch routes, injection/extrusion trials; colorimetry.*
- *Durability & recyclability: thermal/UV aging, reprocessing loops; mechanical and aesthetic retention.*
- *LCA/ESG and regulatory: RoHS/WEEE, K-REACH dossiers.*

Key tasks

- *T1: Synthesis/selection of mono-/oligo-/polymeric and P-functional oxyimides; stability and color screens.*
- *T2: Formulation with Al/Zn phosphinates; dispersion and anti-drip tuning; low-emission delivery forms.*
- *T3: Fire performance validation (UL/LOI/cone); mechanics/aging; migration/VOC/fog compliance.*
- *T4: Scale-up via masterbatch; processing window maps; reprocessing stability; cost/performance models.*
- *T5: Application demos (E&E housings, white goods, automotive interior); conformity testing.*
- *T6: Tech transfer/IP and commercialization roadmap; LCA/ESG dossier.*

Milestones & timeline (3 years)

Phase	Months	Milestone	Description
I	0–2	M1	Kick-off; target applications, compliance criteria, test plans
I	0–6	M2	Candidate oxyimide set (incl. P-functional) synthesized/ selected; Top-10 formulations meeting UL/LOI screens
II	7–12	M3	Masterbatch routes established; processing window maps; initial migration/VOC/fog pass
II	13–18	M4	V-0 at 1.6 mm in PA/PBT and PC/ABS achieved; PHRR/SEA targets met in cone
II	19–24	M5	Application demo parts produced; durability/reprocessing targets achieved
II	25–30	M6	Certification-ready dataset (UL/IEC/ISO); LCA/ESG and cost model completed
II	31–36	M7	OEM-style validation; tech transfer package; commercialization plan finalized

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Technical contributions:* • *Oxyimide/P-functional oxyimide selection, polymeric/oligomeric low-migration designs; formulation architecture with phosphinates.* • *Fire testing (UL/LOI/cone); analytics (TGA/DSC/TGA-MS/FTIR), rheology; migration/VOC/fog; durability and reprocessing stability.* • *Color/stability control, LCA/ESG, and compliance documentation.* *Research scope:* • *Target polymers: PP, PA6/66, PBT, PC/ABS, ABS/HIPS; masterbatch scale-up support; DOE-based optimization and down-selection.* *Resources provided:* • *Compounding/fire/analytics labs; project management; IP management; data packages for certification.*
Expected R&R of Korean SME Partner	*Technical contributions:* • *Fine-chemical synthesis/scale-up of oxyimide/P-functional oxyimide candidates (as agreed IP/licensing); masterbatch production; compounding and pilot parts.* *Development tasks:* • *Processing window mapping in local equipment; color/VOC/fog/migration tests to domestic protocols; cost and supply-chain modeling.* *Testing/validation:* • *Coordinate UL/IEC/ISO conformity tests with Korean test organizations; provide datasets for Phase I/II gates; application-specific validation.* *Resource commitments:* • *Pilot lines, test capacity, materials procurement; regulatory liaison (K-REACH); participation in tech transfer and commercialization.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Technological achievements:* • *Validated halogen-free, antimony-free oxyimide FR packages (including P-functional variants) meeting UL 94 targets with reduced total loading and significantly lower smoke (SEA) at stable mechanics, low migration, and low VOC/fog.* *Commercialization potential:* • *Drop-in masterbatch products and tailored compounds for E&E/automotive/white goods; scalable supply via Korean SMEs; certification-ready documentation; joint IP and licensing routes.* *Industrial/economic impact:* • *ESG-compliant alternatives to halogenated systems; reduced smoke/corrosivity; improved durability and recyclability; strengthened Korea–Germany SME collaboration and export competitiveness in Advanced Mobility markets.*

6. References

(1) WO2014154636A1: Use of organic oxyimides as flame retardants for plastics.

(2) WO2015180888A1: Oxyimide-containing copolymers/polymers as FRs and multifunctional additives.

(3) WO2016042040A1: Phosphorus-containing oxyimides as flame retardants and stabilizers.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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